

EXP 5: COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS

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Subject: Prompt Engineering

1. Aim (Detailed Explanation)

The purpose of this experiment is to systematically evaluate how prompt design influences AI-generated responses. By comparing naïve (unstructured) prompts with basic (structured) prompts, we aim to understand how clarity, context, and specificity impact three key factors:

- **Quality** → How well-written and organized the response is
- **Accuracy** → Whether the information is correct and relevant
- **Depth** → The level of detail and explanation provided

This helps identify best practices for effective prompt engineering.

2. Definition of Prompt Types (Detailed)

Naïve Prompts

Naïve prompts are **simple and vague instructions** with minimal context.

- They rely heavily on the AI's assumptions
- They often lead to **generic or broad responses**
- Example: *"Write a story"*

Limitation: Lack of direction can reduce response quality

Basic Prompts

Basic prompts are **structured and detailed**, guiding the AI clearly.

- Include constraints (word limit, format, focus)
- Provide context and expectations
- Example:
"Write a 150-word story about a boy discovering a hidden power with a clear beginning, conflict, and resolution."

Benefit: Produces **focused and high-quality outputs**

3. Scenario Explanations

Scenario 1: Creative Story

- **Naïve Output Behaviour:**
The AI generates a **general story**, often lacking structure or creativity depth.

- **Basic Output Behaviour:**

The AI follows instructions and includes:

- Beginning
- Conflict

Scenario 2: Factual Question

Naïve Output Behaviour:

Provides a **short and basic definition** of AI.

- **Basic Output Behaviour:**
Includes:
 - Definition
 - Applications
 - Real-world examples
 - **Insight:** Adding constraints improves **completeness and accuracy.**

- **Scenario 3: Summarization**

Naïve Output Behaviour:

May produce:

- Very short summaries
- Missing key points

Basic Output Behaviour:

Ensures:

- Key concepts included
 - Proper length
 - Technical tone
 - **Insight:** Structured prompts improve **precision and relevance**.
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- **Scenario 4: Recommendation**

Naïve Output Behaviour:

Gives **generic suggestions** without reasoning.

Basic Output Behaviour:

Provides:

- Specific recommendations
 - Justification (battery, performance, price)
 - **Insight:** Context leads to **personalized recommendations**.
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- **4. Comparison Table (Detailed Insight)**

- The table highlights how prompt types affect performance:

Naïve Prompts:

- Lower quality
- Less detailed

- Sometimes incomplete

Basic Prompts:

- More structured
- More informative
- Better aligned with user needs

This shows that **prompt design directly impacts output effectiveness.**

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5. Evaluation Criteria (Expanded)

Quality

- Measures clarity, organization, and readability
- Improved by structured instructions

Accuracy

- Measures correctness of information
- Improved by adding context and constraints

Depth

- Measures level of explanation and detail
- Increased when prompts request reasoning or elaboration

6. Analysis (Deep Insight)

- Basic prompts reduce ambiguity and guide the AI toward **specific goals**
- Naïve prompts rely on AI assumptions, leading to **inconsistent results**
- The clearer and more detailed **the prompt**, the better the output

This demonstrates the importance of **prompt engineering in AI usage**

7. Conclusion (Expanded)

- The experiment confirms that:
 - ✓ **Basic prompts consistently outperform naïve prompts**
 - ✓ Structured instructions improve:
 - Clarity
 - Relevance
 - Completeness
 - ✓ Naïve prompts are only effective for:
 - Simple queries
 - Informal or quick responses

8. Final Insight (Very Important for Exams)

- Prompt design acts as a **communication bridge between humans and AI.**

A well-crafted prompt ensures that the AI:

- Understands the task clearly
 - Produces accurate results
 - Delivers meaningful and high-quality responses
 - Therefore, **better prompts = better AI performance**
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